IS7034 – KO_MED Hospital Project

Database Design Document

Deliverables 1, 2 & 3

KO_MED Hospital — Geriatric Care (Ages 60+)

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Course: IS7034

DELIVERABLE 1

Relations, Attributes & Relationships

Part 1: Relations and Their Attributes

The table below identifies each relation (entity) in the KO_MED Hospital database along with its attributes. Primary Keys (PK) and Foreign Keys (FK) are noted.

WARD

Attribute	Key Type	Notes
WardNumber	**PK**	
WardName		
Location		
NumberOfBeds		
TelephoneExtension		

NURSE

Attribute	Key Type	Notes
StaffNumber	**PK**	
FirstName		
LastName		
Gender		
DateOfBirth		
Address		
Telephone		
CurrentTitle		
CurrentSalary		
Shift		
WardNumber	**FK**	References WARD

PCP

Attribute	Key Type	Notes
PracticeNumber	**PK**	
PCPName		
Address		

Attribute	Key Type	Notes
Telephone		

PATIENT

Attribute	Key Type	Notes
PatientNumber	**PK**	
FirstName		
LastName		
Gender		
DateOfBirth		
MaritalStatus		
Address		
Telephone		
InsuranceProvider		
InsuranceNumber		
DateRegistered		
PracticeNumber	**FK**	References PCP

APPOINTMENT

Attribute	Key Type	Notes
AppointmentNumber	**PK**	
AppointmentDate		
AppointmentTime		
ExaminationRoomNumber		
PatientNumber	**FK**	References PATIENT
StaffNumber	**FK**	References NURSE

BED

Attribute	Key Type	Notes
BedNumber	**PK**	
WardNumber	**FK**	References WARD

INPATIENT

Attribute	Key Type	Notes
InpatientID	**PK**	
PatientNumber	**FK**	References PATIENT
BedNumber	**FK**	References BED
WardNumber	**FK**	References WARD
StaffNumber	**FK**	References NURSE
DateWaitlisted		
DatePlaced		
ExpectedStayDays		
DateExpectedLeave		
DateActualDeparture		

DRUG

Attribute	Key Type	Notes
DrugNumber	**PK**	
DrugName		
Description		

MEDICATION

Attribute	Key Type	Notes
AdminID	**PK**	
PatientNumber	**FK**	References PATIENT
DrugNumber	**FK**	References DRUG
DosagePerAdmin		
MethodOfAdmin		
DateOfAdmin		
TimeOfAdmin		

Part 2: Relationships Between Relations

The table below describes the relationships between each pair of related entities.

Relation (Left)	Relationship	Relation (Right)
WARD	has many	**NURSE**
WARD	contains many	**BED**
WARD	has many	**INPATIENT**
PCP	refers many	**PATIENT**
PATIENT	schedules many	**APPOINTMENT**
NURSE	performs many	**APPOINTMENT**
NURSE	allocates bed for	**INPATIENT**
PATIENT	has many	**INPATIENT**
BED	is assigned in	**INPATIENT**
PATIENT	receives many	**MEDICATION**
DRUG	is prescribed in many	**MEDICATION**

Part 3: Assumptions

The following assumptions were made when identifying relations, attributes, and relationships:

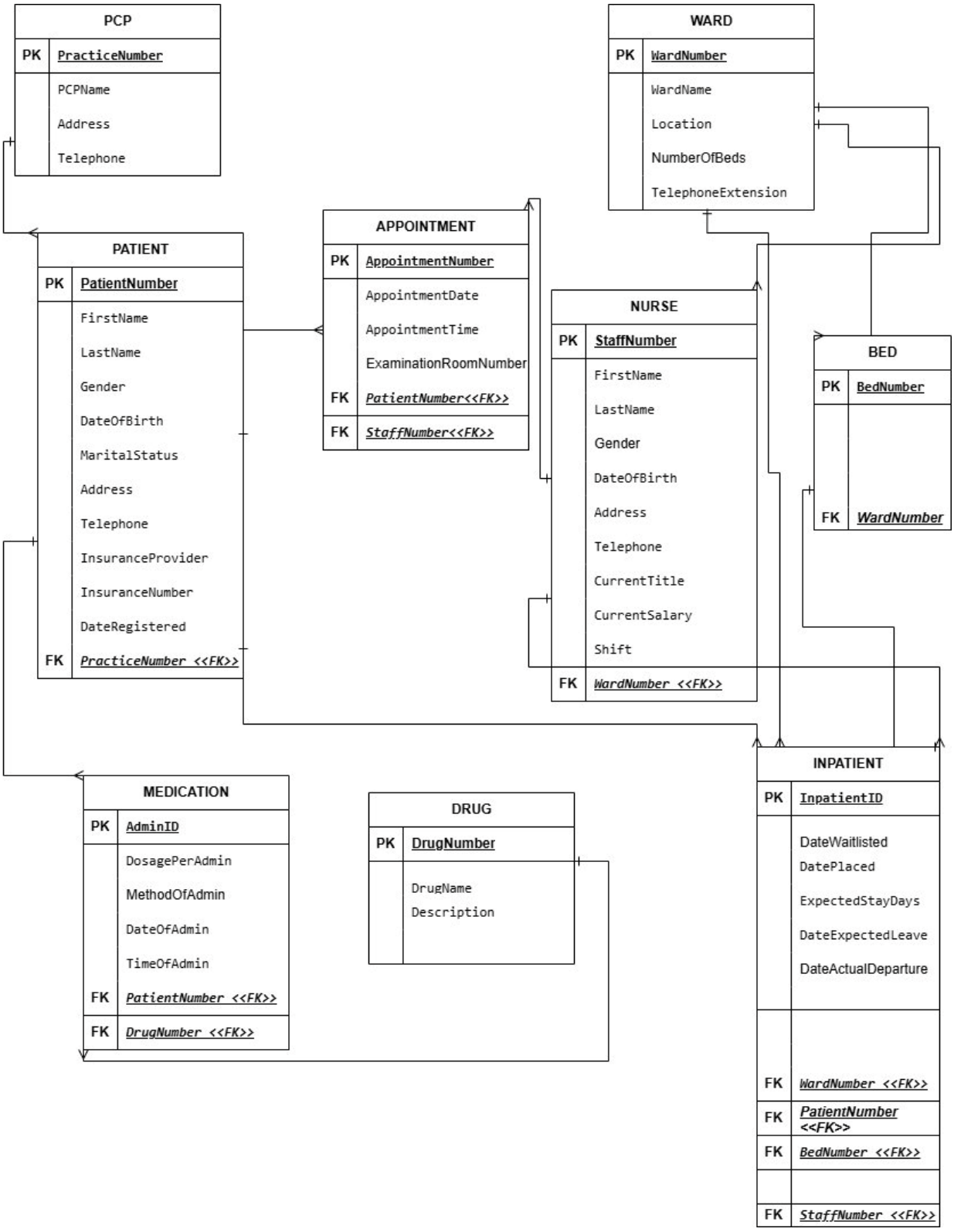
#	Assumption
1	Each nurse is assigned to one ward at a time, though a ward can have many nurses across different shifts (A, B, C).
2	Each bed belongs to exactly one ward and can only be occupied by one active patient at a time.
3	A patient has exactly one PCP at the time of registration, but a PCP can refer many patients.
4	An appointment is performed by one nurse, though a nurse can perform many appointments over time.
5	Only inpatients are tracked in this system — outpatients are explicitly excluded per the case description.
6	A patient can have multiple inpatient stays over time (e.g., readmissions), each recorded as a separate INPATIENT record.
7	BED is modeled as its own relation because beds have unique numbers and belong to specific wards.
8	DRUG is separated from MEDICATION because the same drug can be administered to multiple patients — storing drug details separately avoids redundancy.
9	The nurse who performed the examination is also responsible for the inpatient bed allocation, consistent with the case description.
10	MaritalStatus is limited to two values: Married or Single, as specified in Appendix 2 of the case study.

DELIVERABLE 2

Entity Relationship Diagram

ER Diagram — KO_MED Hospital

The diagram below shows all 9 entities, their attributes, primary keys (PK), foreign keys (FK), and the relationships between them using Crow's Foot notation.



Tool used: diagrams.net (draw.io) | Notation: Crow's Foot (IE Notation) | All relationships show cardinality (1 to many or 1 to 1).

DELIVERABLE 3

Normalization: UNF → 1NF → 2NF → 3NF

Overview of Normalization

Normalization is the process of organizing a relational database to reduce data redundancy and improve data integrity. It involves decomposing a large flat table into smaller well-structured relations by applying a series of rules called Normal Forms. This document walks through each step — Unnormalized Form (UNF), First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF) — applied to the KO_MED Hospital data.

Step 1: Unnormalized Form (UNF)

What is UNF?

Unnormalized Form is the raw flat representation of data exactly as it appears in the source documents prior to applying any normalization rules. In UNF a relation may contain repeating groups, non-atomic values, and no clear primary key.

Problems Identified in KO_MED UNF Data

Problem 1 — Repeating Groups: A single patient (e.g., P1234) can have multiple drug administrations. In UNF these are jammed into one cell: 'Penicillin 1mg IV; Advil 15mg Oral' — this violates atomicity.

Problem 2 — No Primary Key: There is no single field that uniquely identifies every row.

Problem 3 — Mixed Entities: Ward, Patient, and Medication information are all mixed into one flat structure.

Problem 4 — Null Values: Patients on the waiting list have no DatePlaced or DateActualDeparture values yet.

> UNF Rule: No rules applied yet. Data is in its raw state from the source documents — Appendix 4 (Patient Assignment) and Appendix 5 (Patient Medication Record).

UNF Table — Combined Patient Assignment + Medication (Raw/Flat)

BedNumber	PatientNumber	Patient Name	WardNumber	Ward Name	DateWaitlisted	DatePlaced	Exp Stay	DateExpLeave	Drug/Dosage (REPEATING)
O5	P1234	Patient1	W01	Orthopedic	09/26/15	09/30/15	5	10/05/15	**Penicillin 1mg IV; Advil 15mg Oral**
O3	P0782	Patient2	W01	Orthopedic	10/01/15	10/03/15	7	10/10/15	**Advil 15mg Oral**

Step 2: First Normal Form (1NF)

What is 1NF?

A relation is in First Normal Form when every cell contains a single atomic value, all values in a column are the same data type, every column has a unique name, a primary key is defined, and no duplicate rows exist.

Actions Taken to Achieve 1NF

Action 1 — Separate Medication repeating group: Each drug administration is placed on its own row in a separate MEDICATION table with composite PK (PatientNumber, DrugNumber).

Action 2 — Assign composite PK to Patient Assignment: Composite key of (BedNumber, PatientNumber) used.

Action 3 — Ensure atomic columns: All date, name, and numeric fields are atomic — no concatenated values remain.

Action 4 — Remove duplicates: Each row is now uniquely identifiable.

> 1NF Achieved: All cells are atomic. Composite key defined. Repeating groups eliminated by separating medication into its own table.

1NF Table A — Patient Assignment (Composite PK: BedNumber + PatientNumber)

BedNum(PK)	PatientNum(PK)	Patient Name	Ward Num	DateWaitlisted	DatePlaced	Exp Stay	DateExpLeave	DateActualDep	Ward Name
O5	**P1234**	Patient1	W01	09/26/15	09/30/15	5	10/05/15	10/07/15	Orthopedic
O3	**P0782**	Patient2	W01	10/01/15	10/03/15	7	10/10/15	—	Orthopedic

1NF Table B — Medication (Composite PK: PatientNumber + DrugNumber)

PatientNum(PK)	DrugNumber(PK)	DrugName	Description	Dosage	Method	Date & Time
P1234	**234567**	Penicillins	Antibiotic	1mg/ml	IV	10/01/15 08:45
P1234	**023481**	Advil	Pain Killer	15mg/ml	Oral	10/02/15 17:26

Step 3: Second Normal Form (2NF)

What is 2NF?

A relation is in 2NF when it is in 1NF AND every non-key attribute is fully functionally dependent on the entire primary key — not just part of it. Partial dependencies must be removed.

Partial Dependencies Found and Removed

PD 1 — WardName: Depends only on WardNumber (part of key), not on PatientNumber. Moved to WARD table.
PD 2 — WardNumber: Depends only on BedNumber. Moved to WARD table.
PD 3 — DrugName: Depends only on DrugNumber, not on PatientNumber. Moved to DRUG table.
PD 4 — Description: Same as DrugName — depends solely on DrugNumber. Moved to DRUG table.

> 2NF Achieved: All partial dependencies removed. Ward attributes moved to WARD table. Drug attributes moved to DRUG table.

2NF — INPATIENT (WardName/WardNumber partial deps removed)

BedNum(PK)	PatientNum(PK)	WardNum(FK)	DateWaitListed	DatePlaced	ExpStay	DateExpLeave	DateActualDep
O5	**P1234**	W01	09/26/15	09/30/15	5	10/05/15	10/07/15

2NF — WARD (Split out — WardName depends only on WardNumber)

WardNum(PK)	WardName	Location	NumBeds	TelExt
W01	Orthopedic	Building C	20	12345

2NF — MEDICATION (DrugName/Description partial deps removed)

PatientNum(PK)	DrugNum(PK)	AdminID(PK)	Dosage	Method	DateAdmin	TimeAdmin
P1234	**234567**	**A001**	1mg/ml	IV	10/01/15	08:45

2NF — DRUG (Split out — DrugName depends only on DrugNumber)

DrugNum(PK)	DrugName	Description
234567	Penicillins	Antibiotic
023481	Advil	Pain Killer

Step 4: Third Normal Form (3NF)

What is 3NF?

A relation is in 3NF when it is in 2NF AND there are no transitive dependencies. A transitive dependency occurs when a non-key attribute depends on another non-key attribute rather than directly on the primary key.

Transitive Dependency Check — All Relations

Relation	Analysis	Result
WARD	WardName, Location, NumberOfBeds, TelephoneExtension all depend directly on WardNumber.	**No transitive dependencies — already in 3NF**
DRUG	DrugName and Description depend directly on DrugNumber.	**No transitive dependencies — already in 3NF**
MEDICATION	All admin fields depend directly on AdminID.	**No transitive dependencies — already in 3NF**
INPATIENT	All date fields depend directly on InpatientID. WardNumber is a proper FK.	**No transitive dependencies — already in 3NF**
PATIENT	All registration fields depend directly on PatientNumber. PracticeNumber is a proper FK.	**No transitive dependencies — already in 3NF**
NURSE	All staff fields depend directly on StaffNumber. WardNumber is a proper FK.	**No transitive dependencies — already in 3NF**
APPOINTMENT	All fields depend directly on AppointmentNumber.	**No transitive dependencies — already in 3NF**
BED	WardNumber depends on BedNumber directly.	**No transitive dependencies — already in 3NF**
PCP	All PCP fields depend directly on PracticeNumber.	**No transitive dependencies — already in 3NF**

3NF Achieved: No transitive dependencies exist in any relation. All 9 final relations are in 3NF. This schema is production-ready for SQL implementation.

Final 3NF Schema — All 9 Relations Summary

Relation	Attributes (PK underlined, FK noted)
WARD	WardNumber (PK), WardName, Location, NumberOfBeds, TelephoneExtension
NURSE	StaffNumber (PK), FirstName, LastName, Gender, DateOfBirth, Address, Telephone, CurrentTitle, CurrentSalary, Shift, WardNumber (FK)
PCP	PracticeNumber (PK), PCPName, Address, Telephone
PATIENT	PatientNumber (PK), FirstName, LastName, Gender, DateOfBirth, MaritalStatus, Address, Telephone, InsuranceProvider, InsuranceNumber, DateRegistered, PracticeNumber (FK)
APPOINTMENT	AppointmentNumber (PK), AppointmentDate, AppointmentTime, ExaminationRoomNumber, PatientNumber (FK), StaffNumber (FK)
BED	BedNumber (PK), WardNumber (FK)
INPATIENT	InpatientID (PK), PatientNumber (FK), BedNumber (FK), WardNumber (FK), StaffNumber (FK), DateWaitlisted, DatePlaced, ExpectedStayDays, DateExpectedLeave, DateActualDeparture
DRUG	DrugNumber (PK), DrugName, Description
MEDICATION	AdminID (PK), PatientNumber (FK), DrugNumber (FK), DosagePerAdmin, MethodOfAdmin, DateOfAdmin, TimeOfAdmin

End of Deliverables 1, 2 & 3 — KO_MED Hospital Database Design